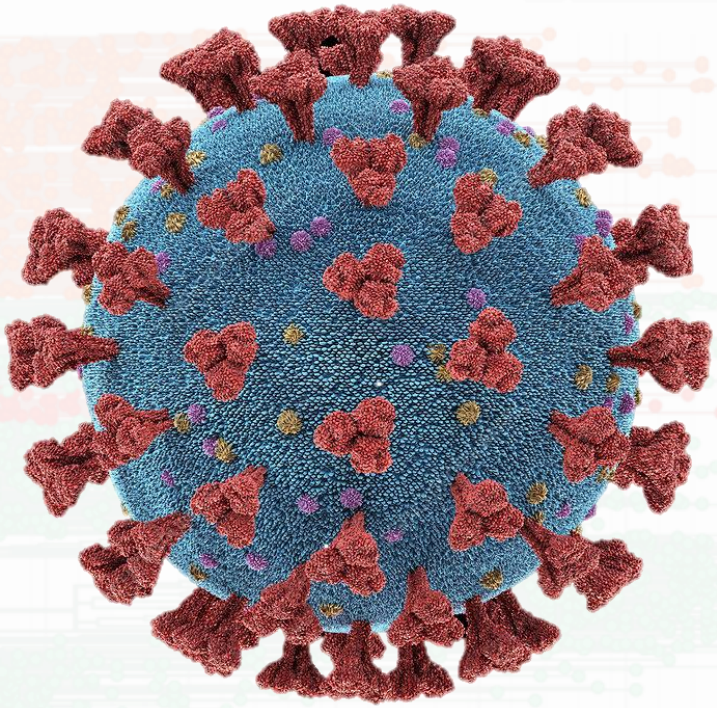
Chasing the coronavirus

a global, collaborative effort to identify the next variant



Julius Winter
Clinical bioinformatics, SIB

Swiss Institute of Bioinformatics

IN THE LIGHT OF EVOLUTION

Swiss National Science Foundation

CC BY NC

Wer bin ich?



Julius Winter

Matura 2012 – Gymnasium Leonhard (Schwerpunkt Spanisch)

Bachelor Nanowissenschaften	Uni Basel
Master Nanowissenschaften	Uni Basel
PhD Bioengineering	ETH Lausanne

Seit 2024 Computational Biologist beim SIB in Genf

From nose to database

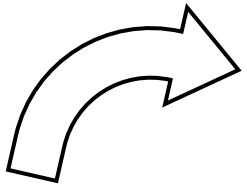


From nose to database

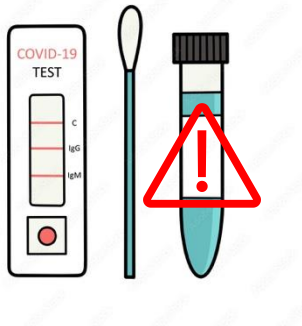




From nose to database

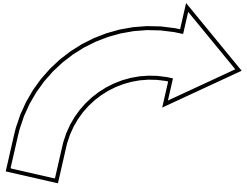




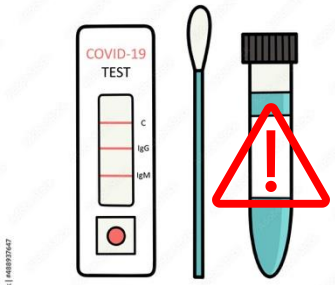




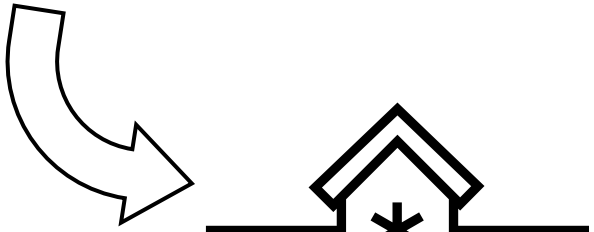
From nose to database

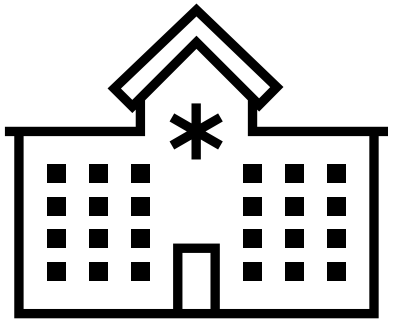








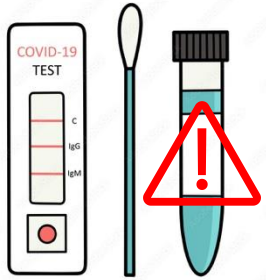




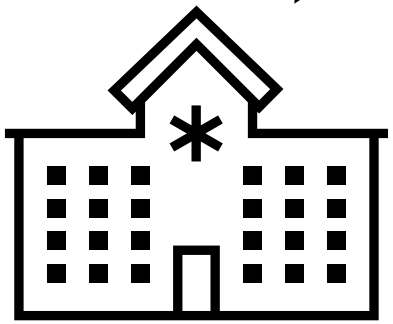
sequencing lab

From nose to database









sequencing lab

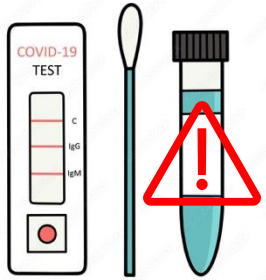
National Library of Medicine
National Center for Biotechnology Information

GISAID

From nose to database







NIH National Library of Medicine
National Center for Biotechnology Information

GISAID

cttgtccctggtttcaacgagaaaac...

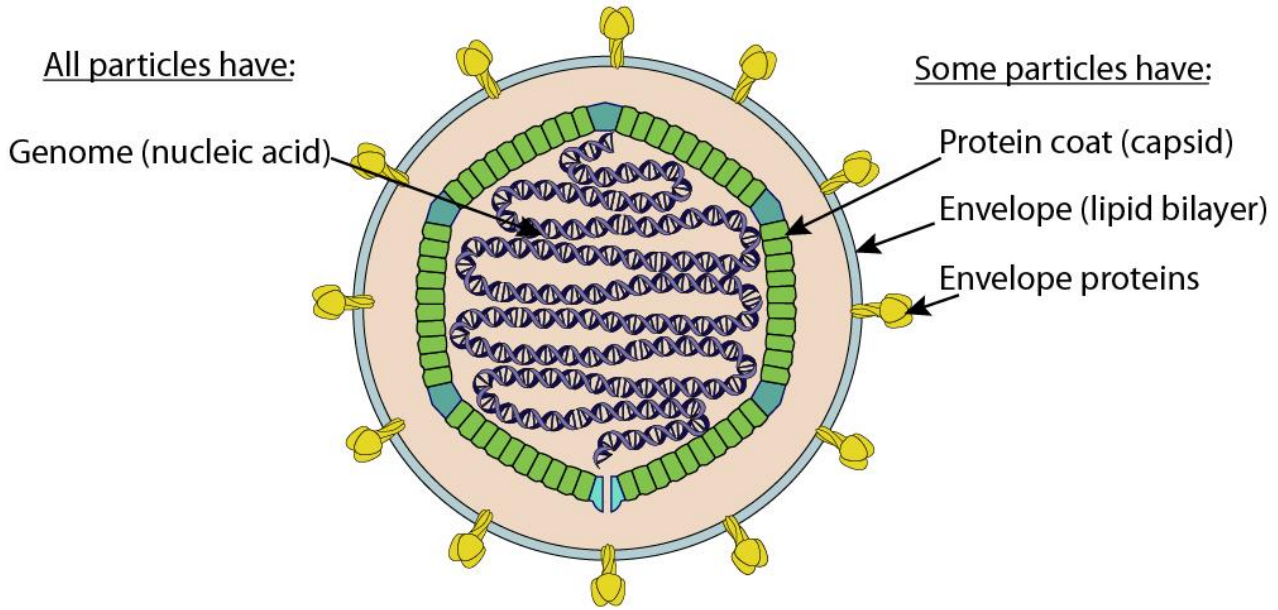
sequencing lab

New variant!

Let’s take a step back – what even is a virus?

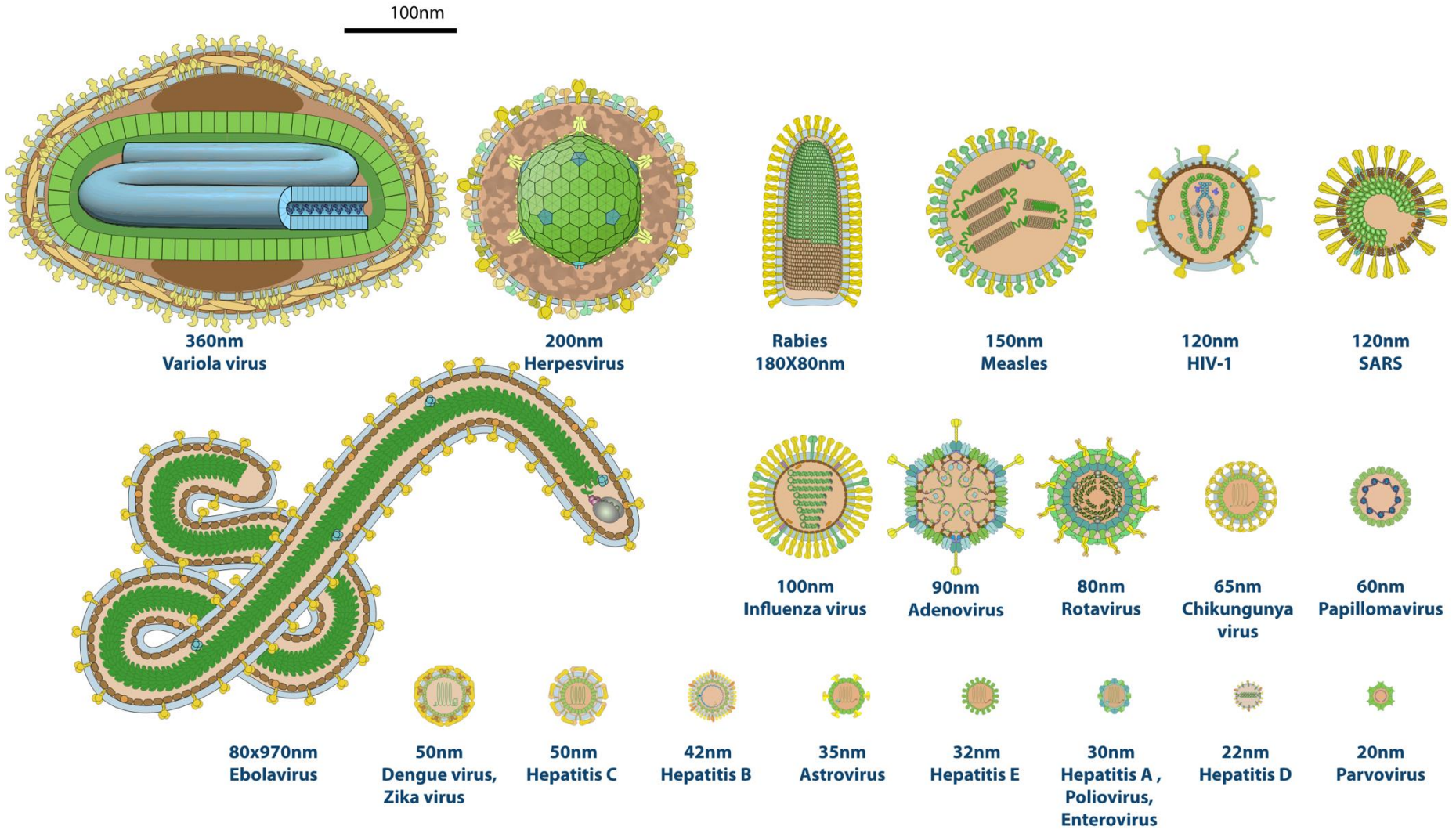
- **building instructions** → genetic material (DNA or RNA)
- **protective box, address label, etc.** → protein shell, surface proteins





They come in all shapes and sizes

Eradicated since 1980 thanks to the variola vaccine*

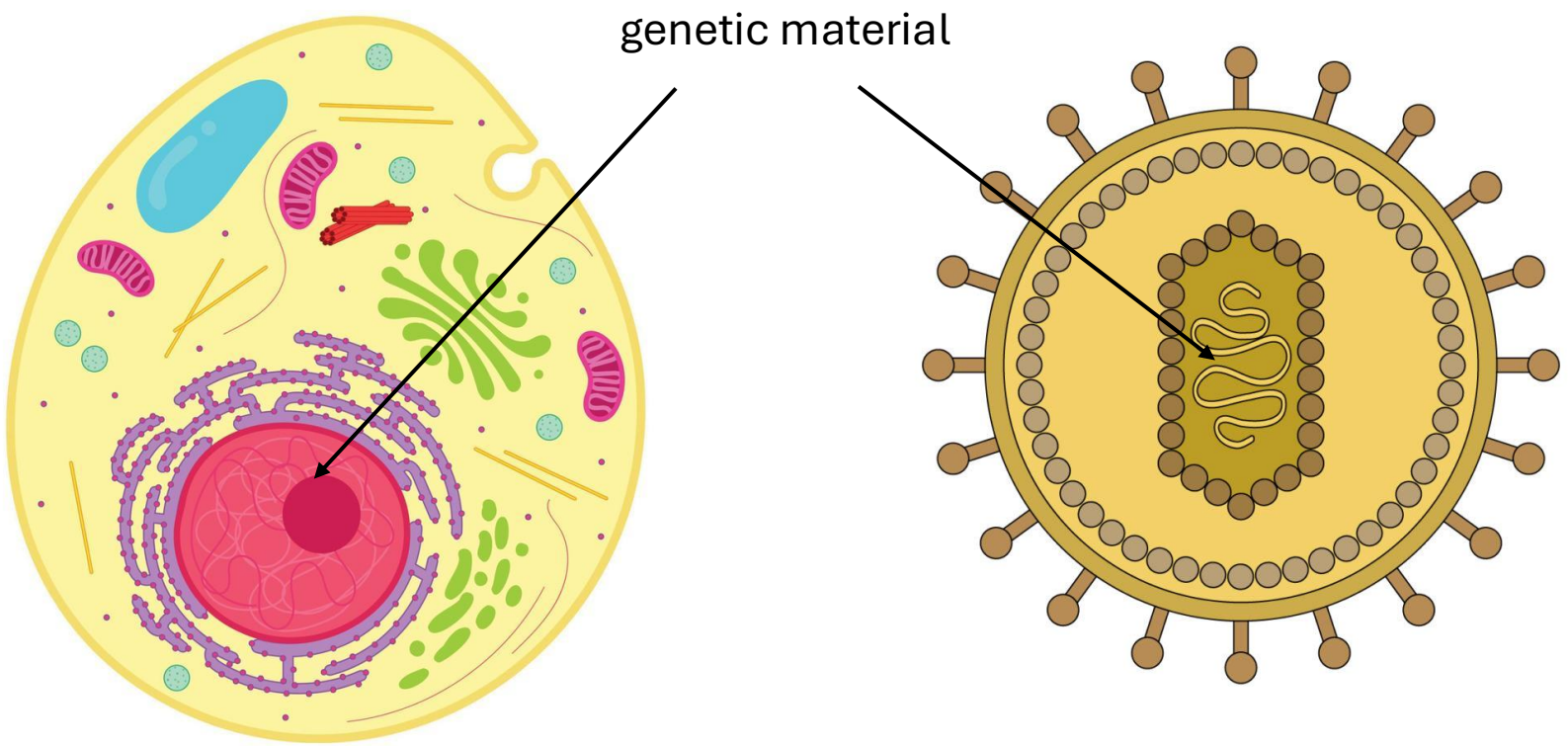


https://viralzone.expasy.org/5216

*https://iris.who.int/items/ba4ab312-1c43-4304-8235-969979499717

What makes a virus dangerous?

Only about ~200 virus types lead to infectious diseases in humans
https://viralzone.expasy.org/678

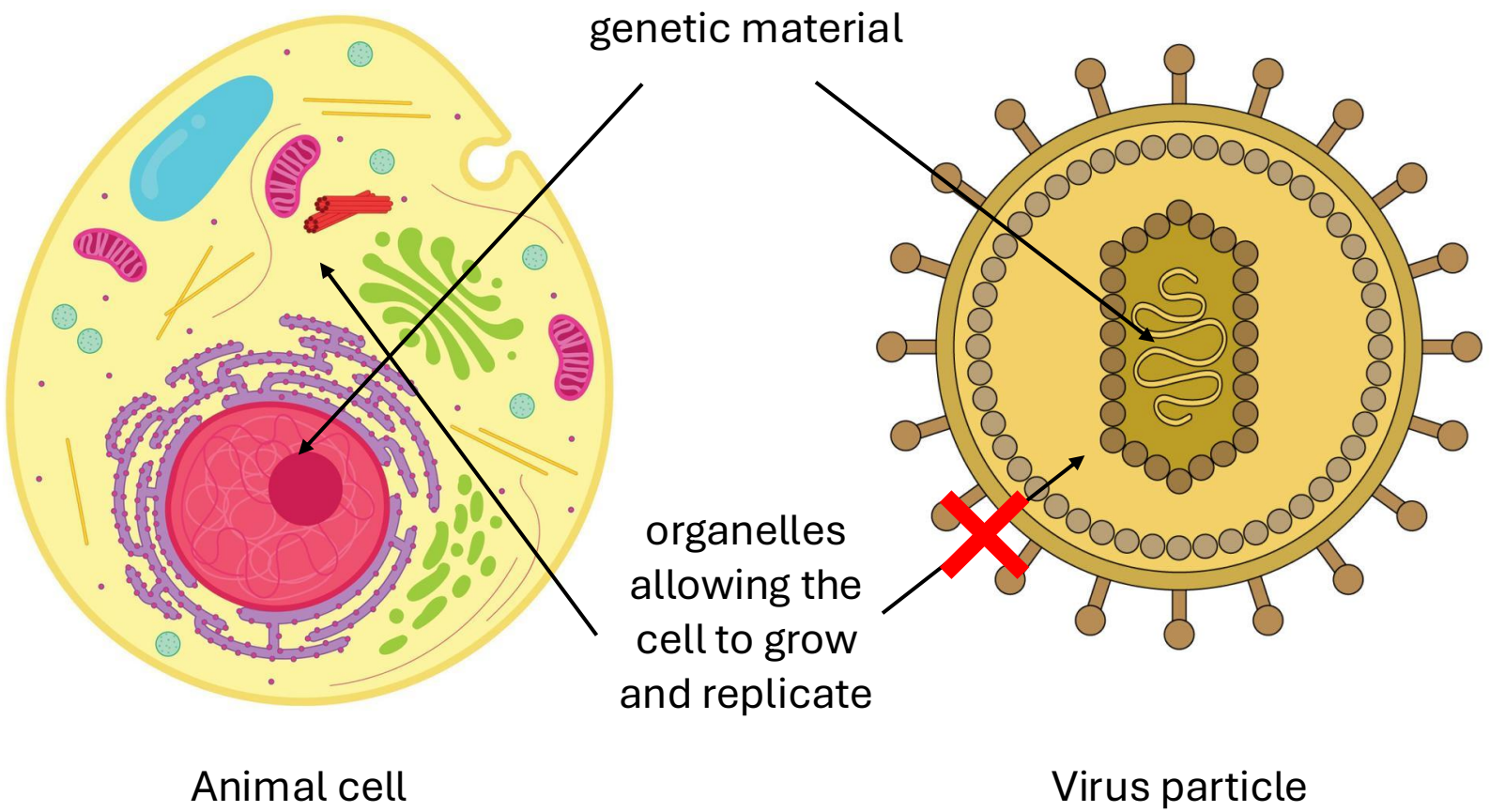


Animal cell

Virus particle

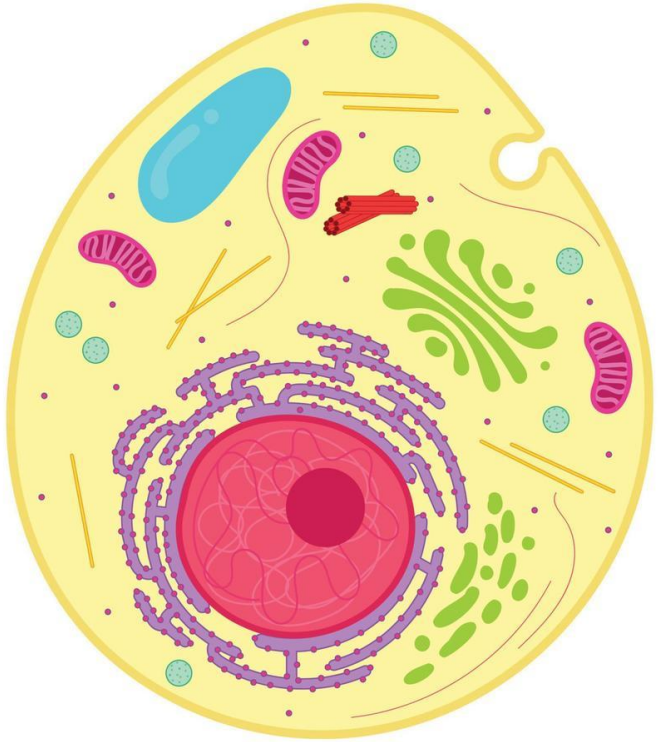
What makes a virus dangerous?

Only about ~200 virus types lead to infectious diseases in humans
https://viralzone.expasy.org/678



What makes a virus dangerous?

Only about ~200 virus types lead to infectious diseases in humans
https://viralzone.expasy.org/678

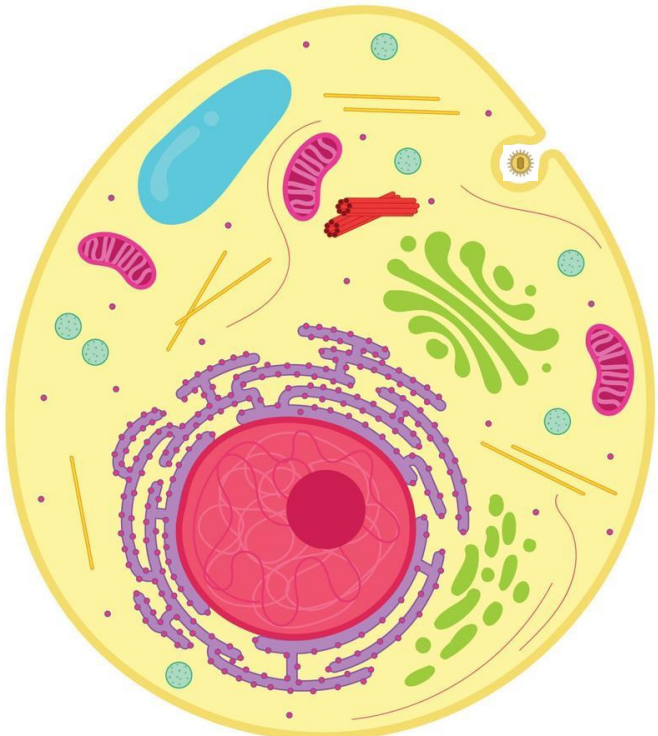


Animal cell

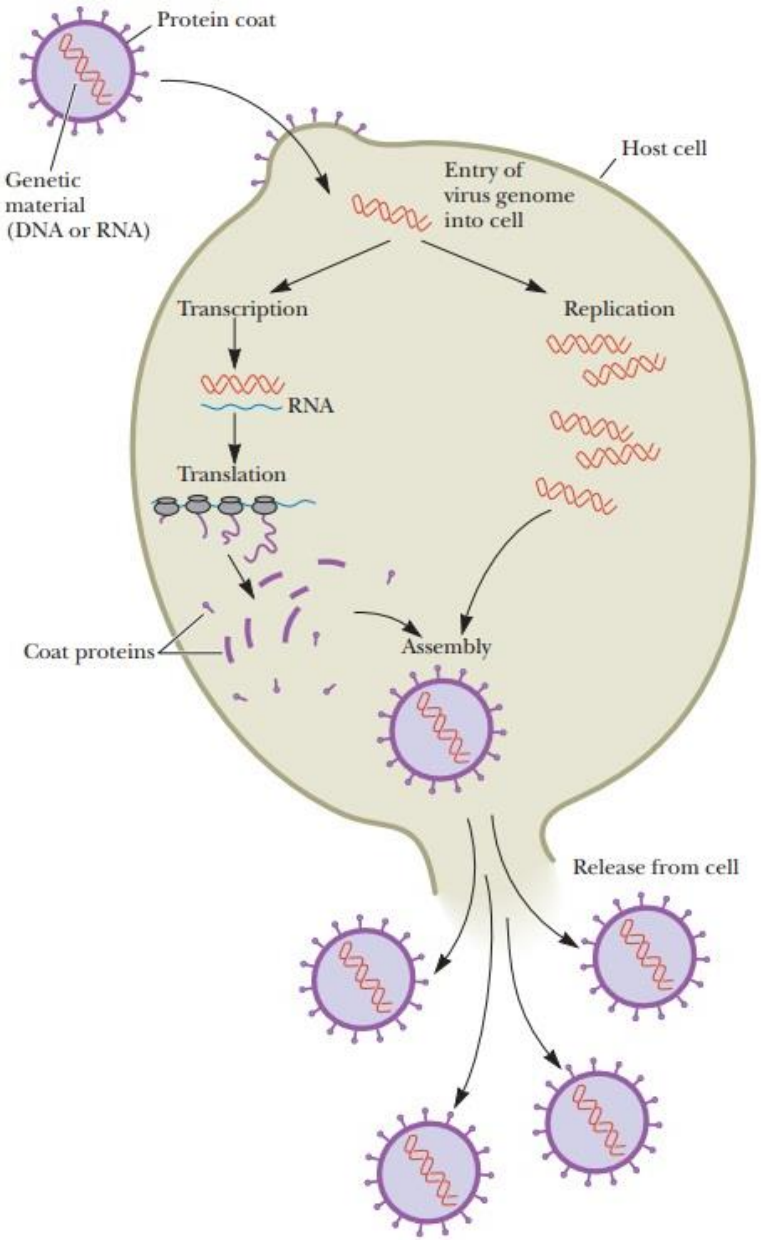
~200-300x smaller

Virus particle

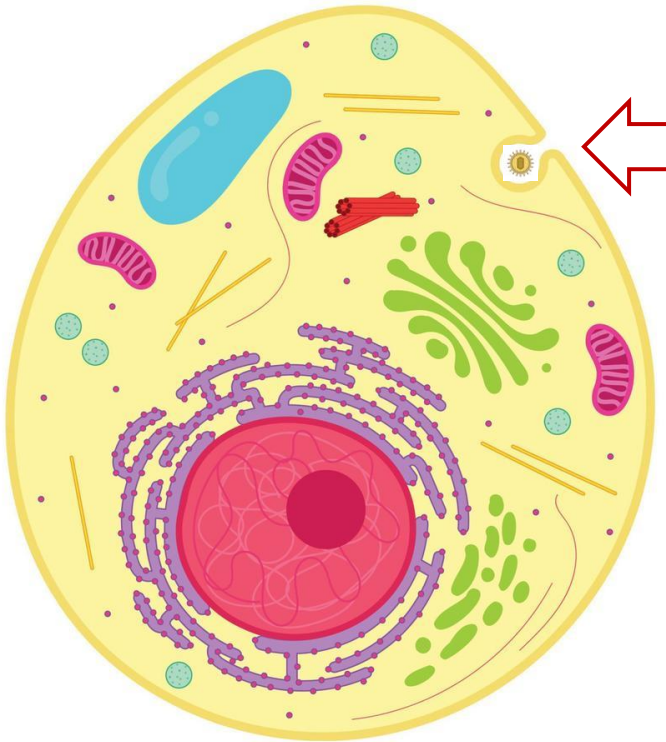
What makes a virus dangerous?



virus enters host cell an hijacks it

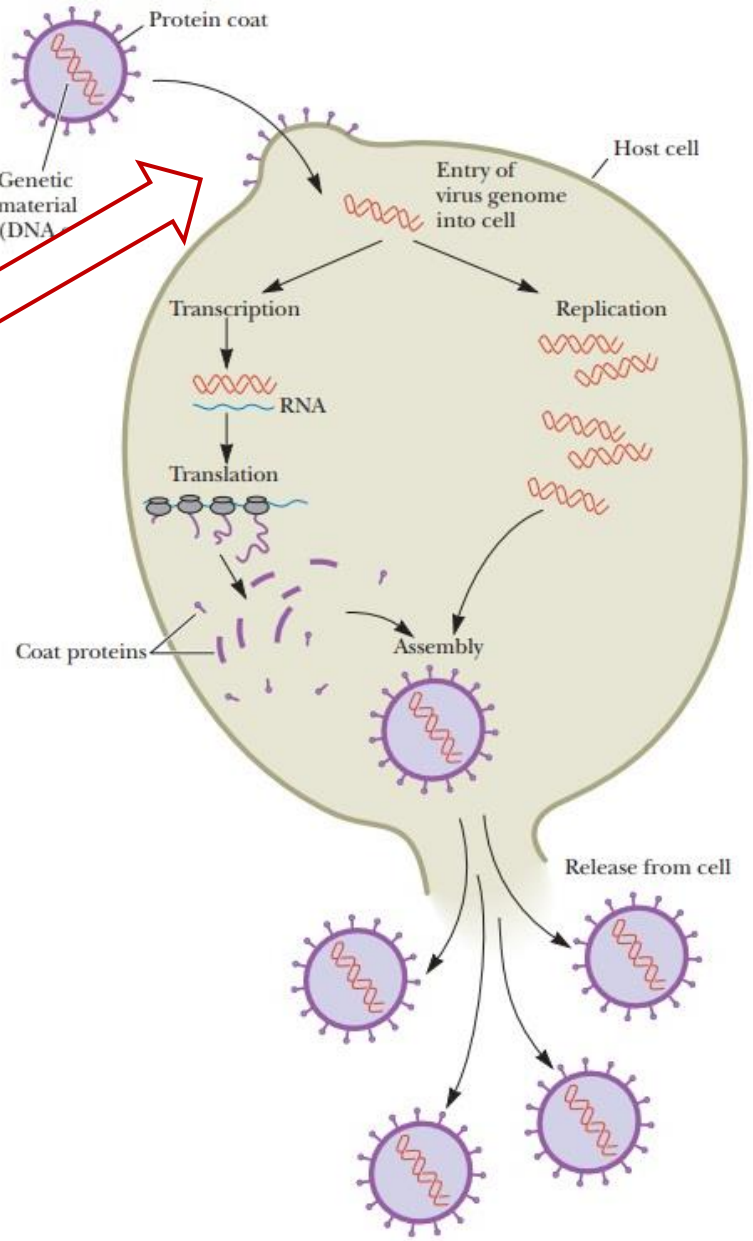


What makes a virus dangerous?

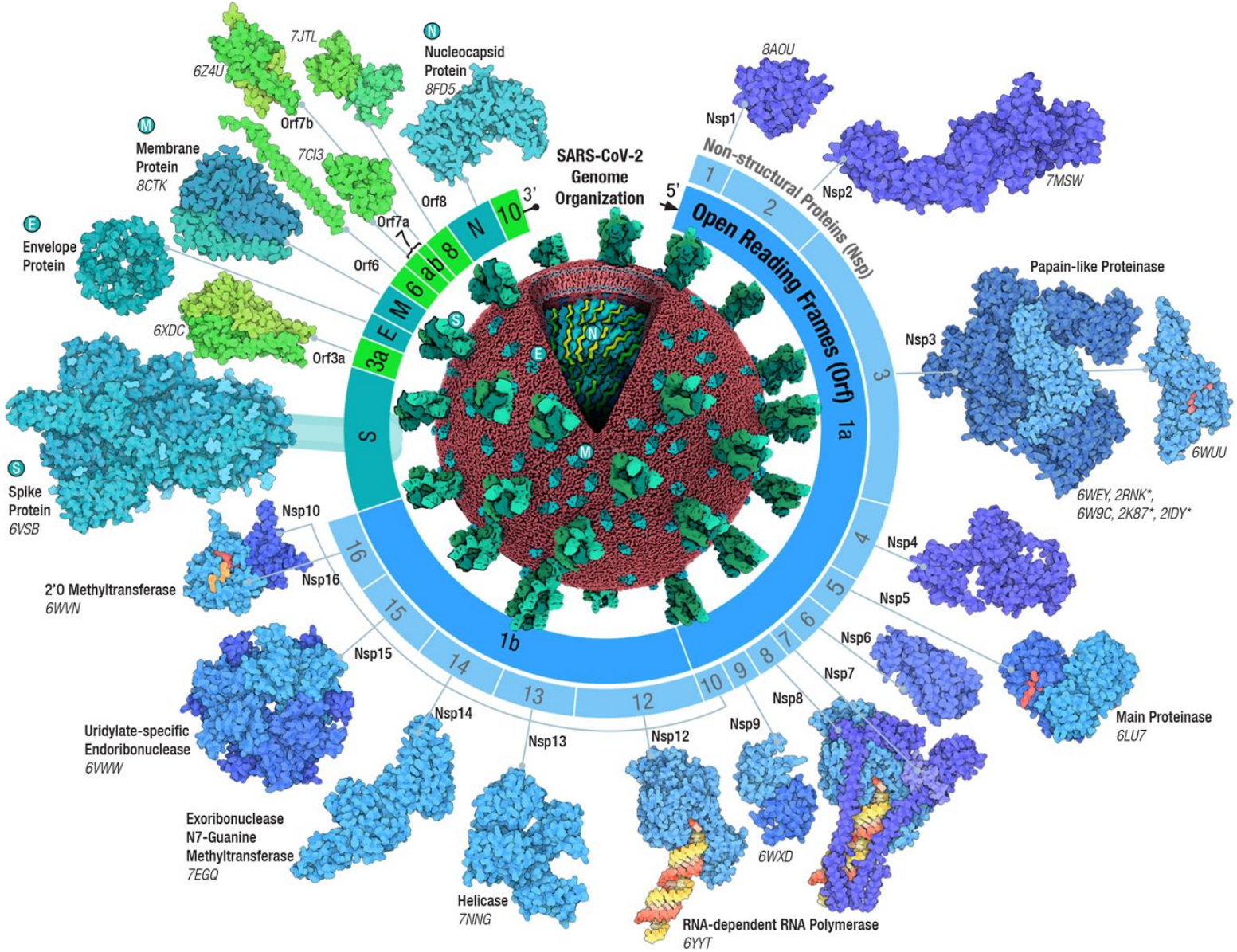


virus enters host cell an hijacks it

one of the most important steps



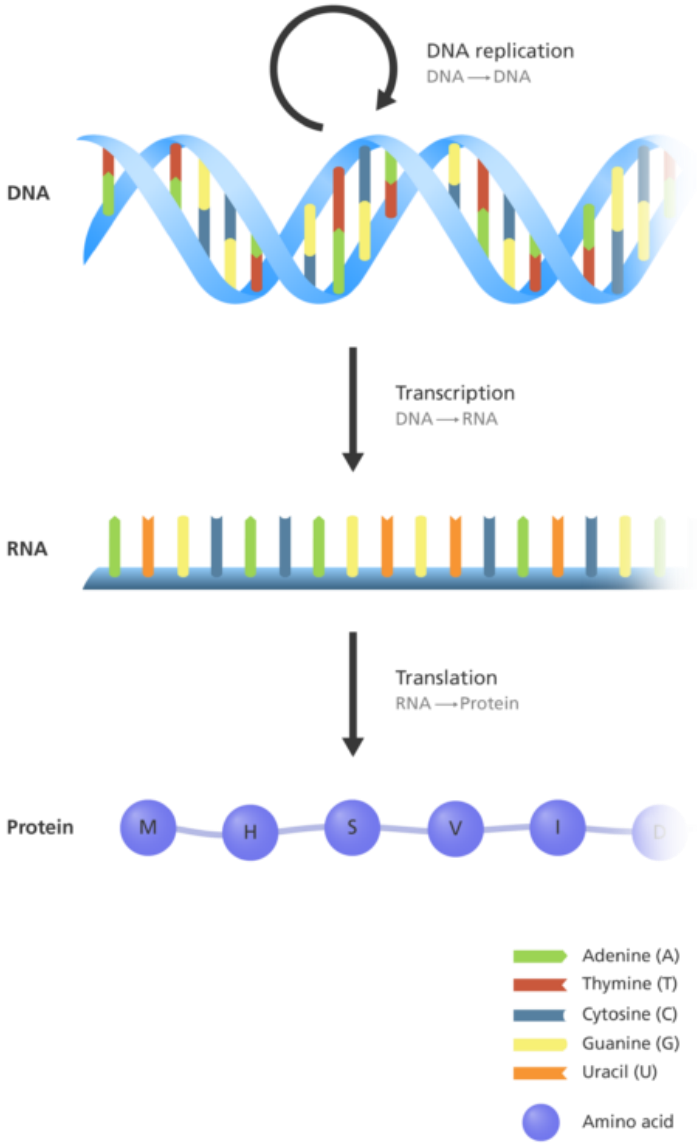
Just a few genes to make it all work



~29’000 bases long

```
>NC_045512.2 Severe acute respiratory syndrome coronavirus 2 isolate Wuhan-Hu-1,
complete genome
ATTAAAGGTTTATACCTTCCCAGGTAACAAACCAACCAACTTTCGATCTCTTGTAGATCTGTTCTCTAAA
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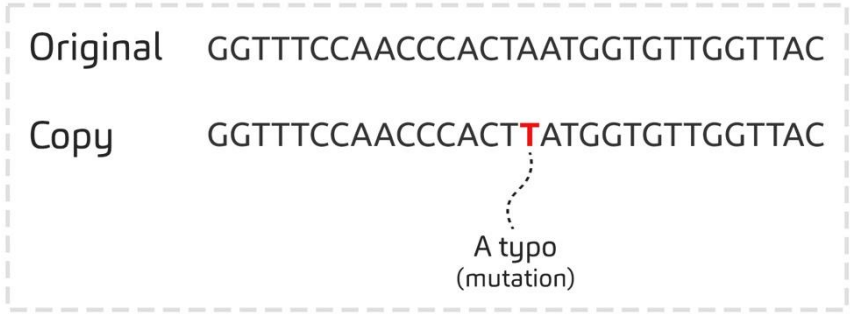
Mistakes can happen during replication



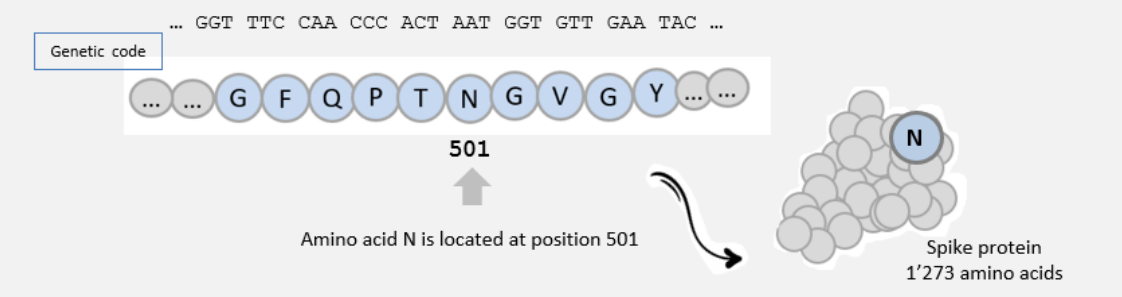
Original	GGTTTCCAACCCACTAATGGTGTTGGTTAC
Copy	GGTTTCCAACCCACTTATGGTGTTGGTTAC

A typo
(mutation)

Mutations change the behavior of the encoded protein



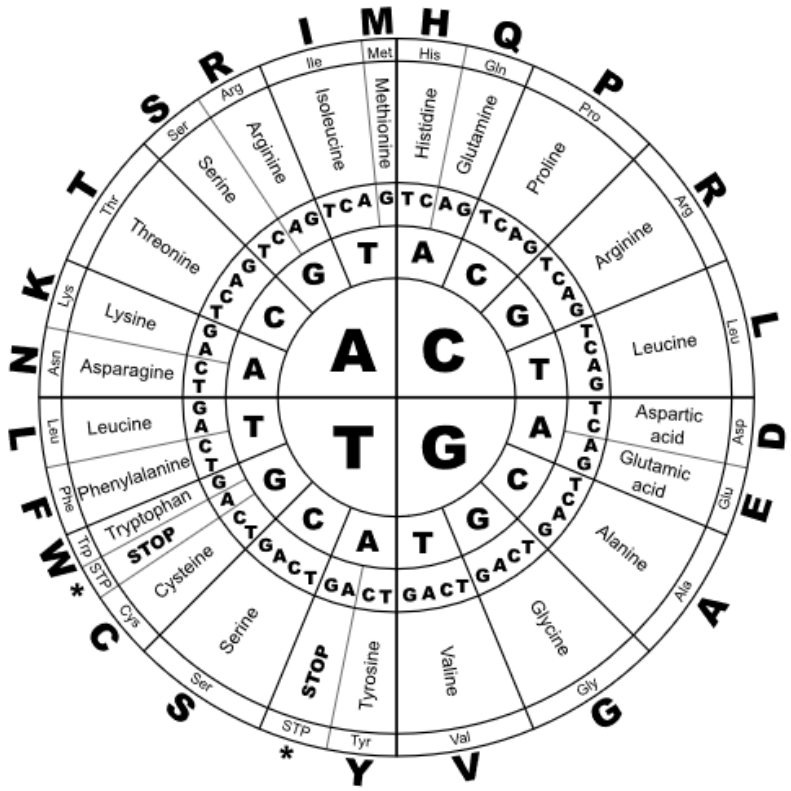
Mutations change the behavior of the encoded protein



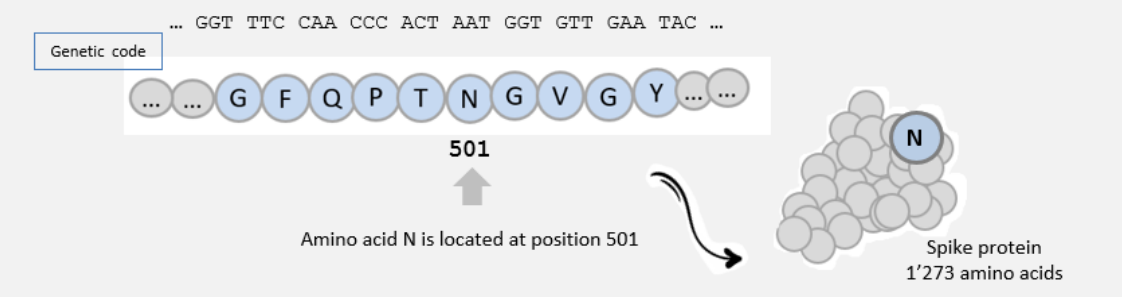
Original GGTTTCCAACCCACTAATGGTGTTGGTTAC

Copy GGTTTCCAACCCACTTATGGTGTTGGTTAC

A typo
(mutation)



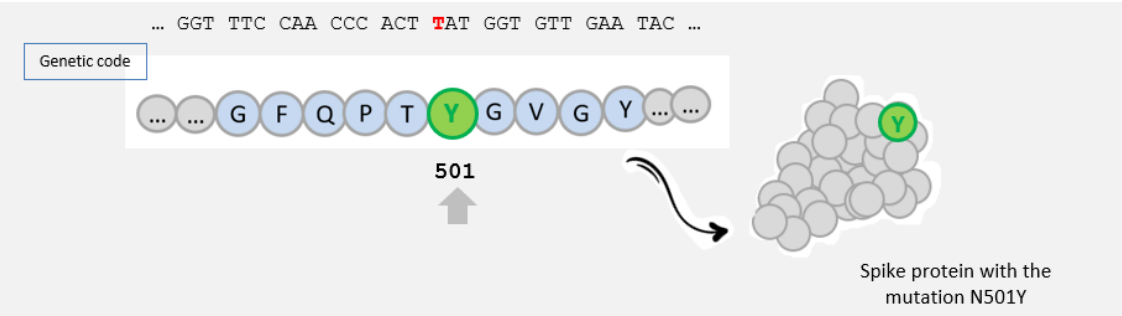
Mutations change the behavior of the encoded protein

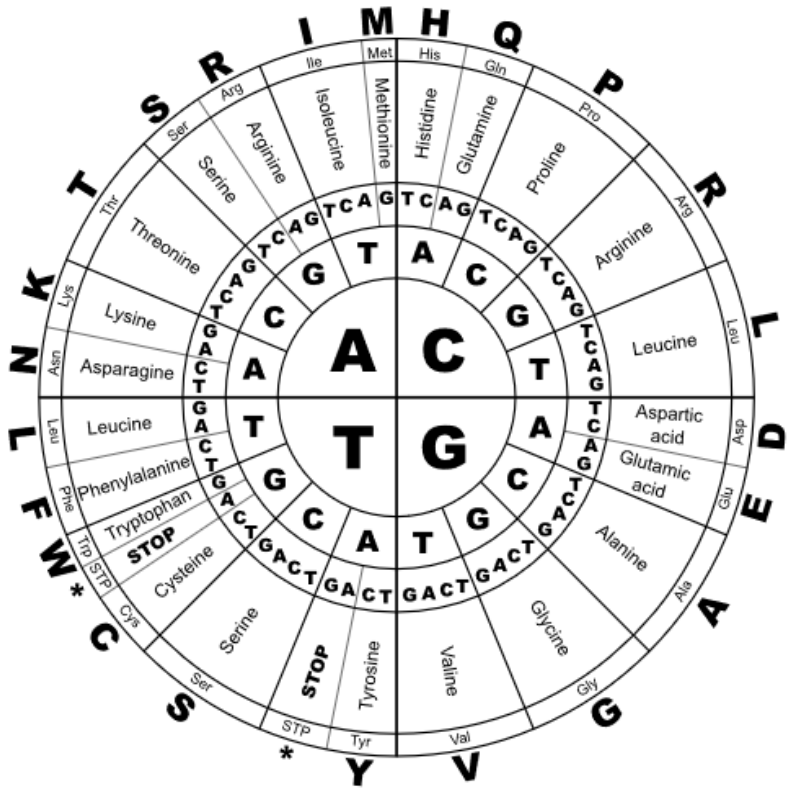


Original GGTTTCCAACCCACTAATGGTGTTGGTTAC

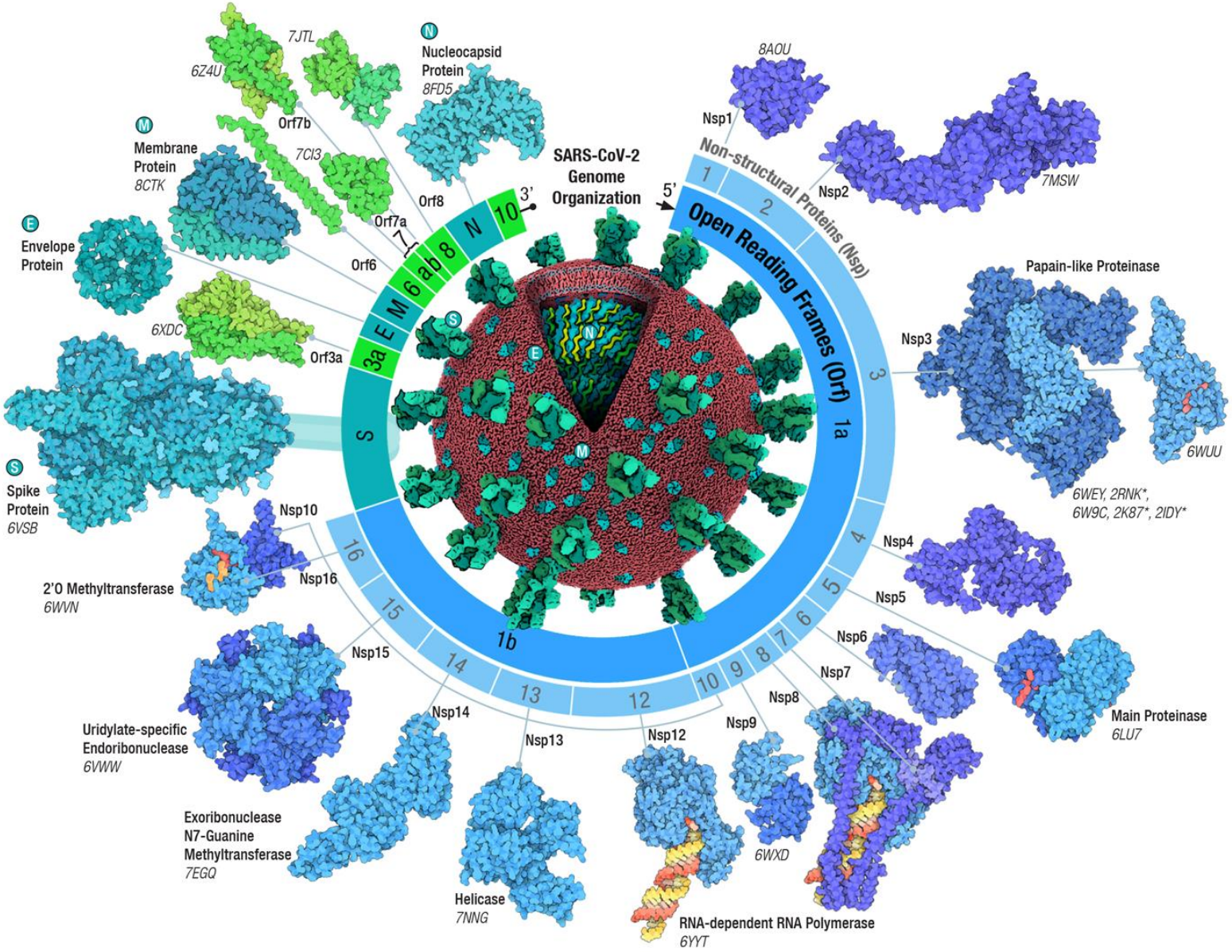
Copy GGTTTCCAACCCACTTATGGTGTTGGTTAC

A typo
(mutation)

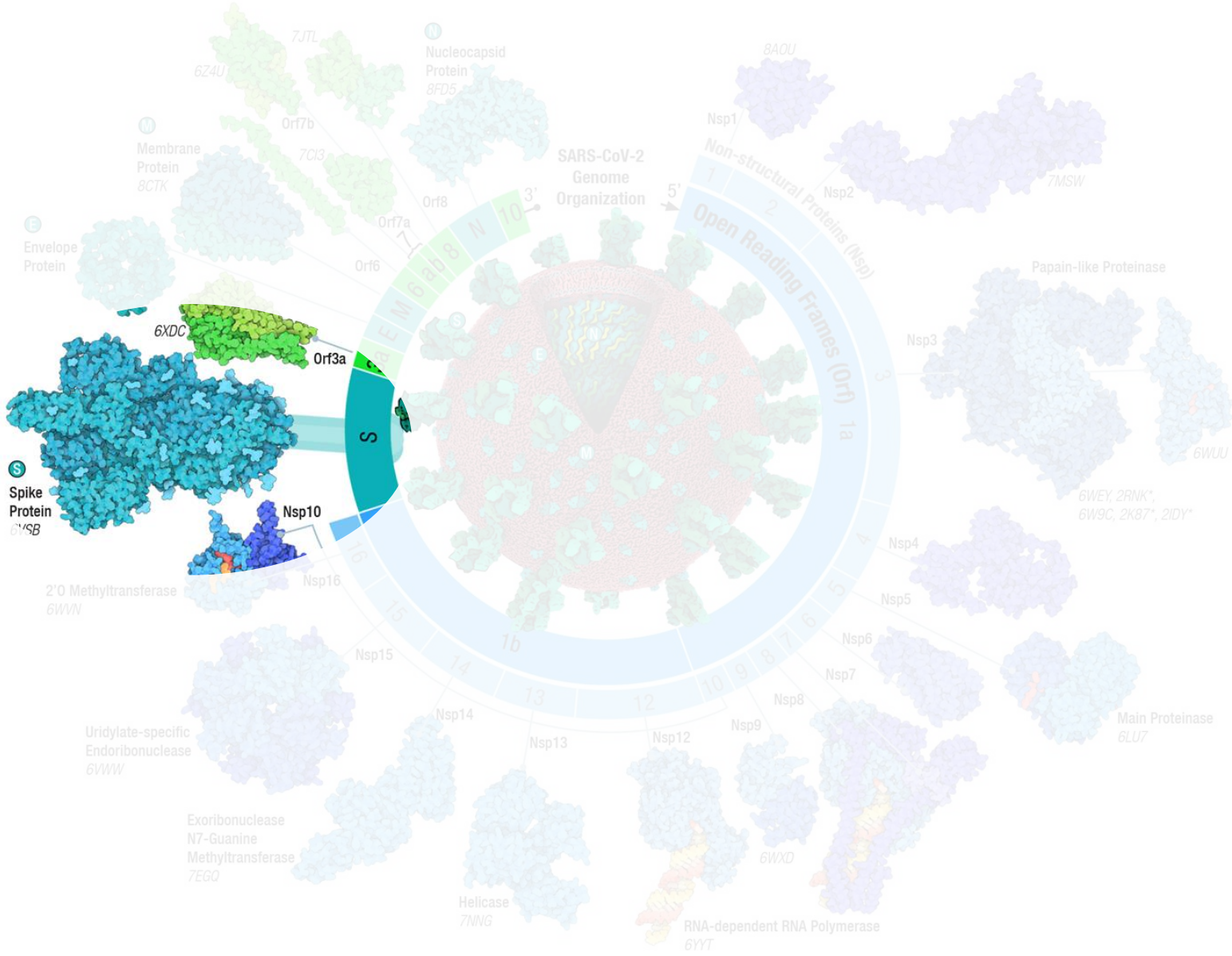




One protein stands out

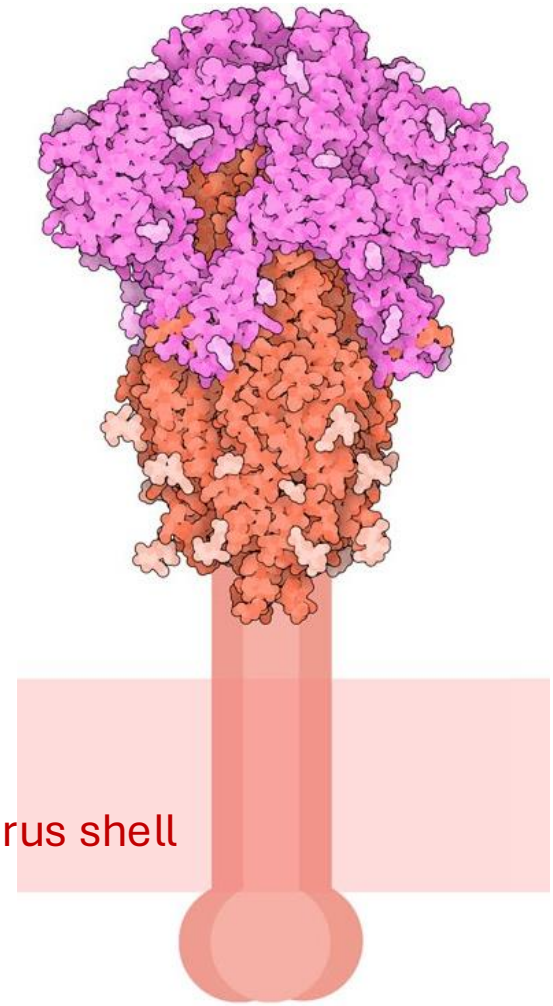


One protein stands out



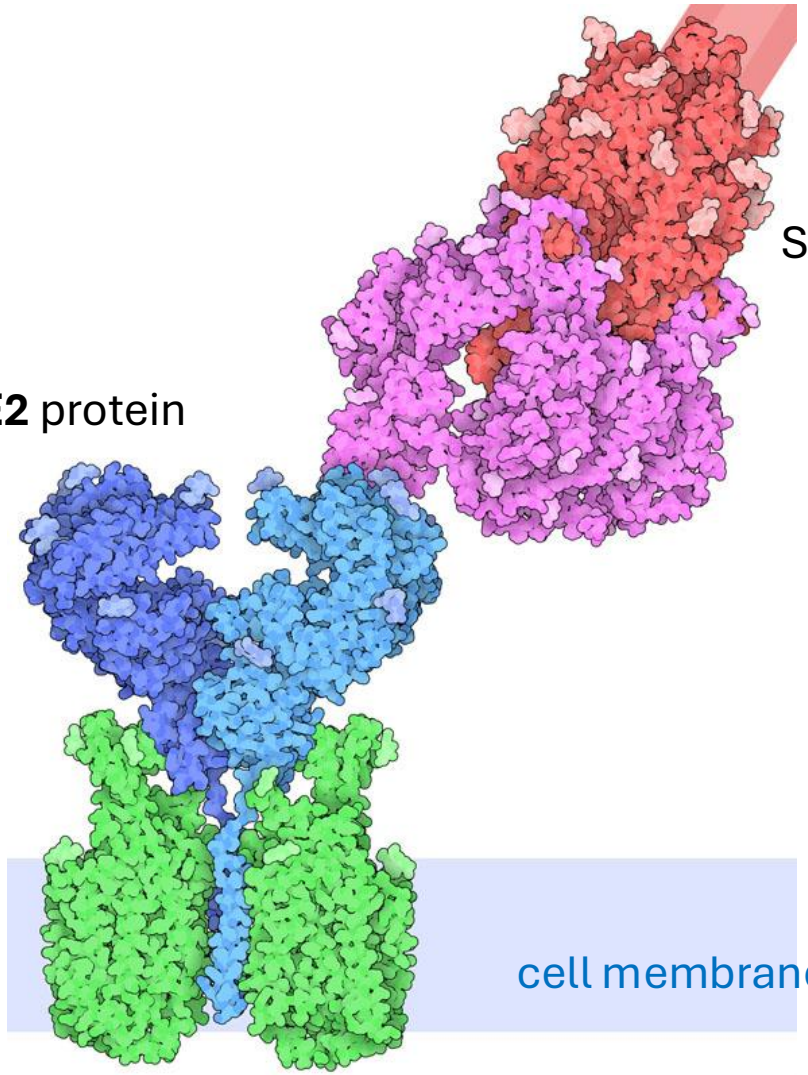
The spike protein is the key into the cell

SARS-CoV-2 **Spike**



virus shell

human **ACE2** protein



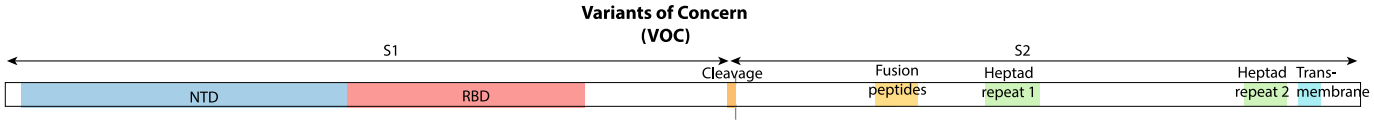
SARS-CoV-2 **Spike**

cell membrane

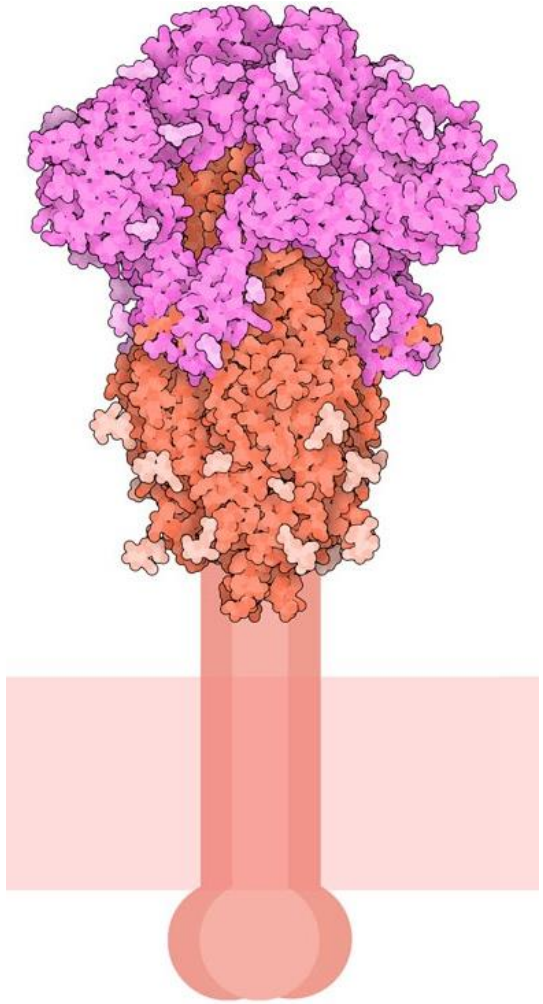
ACE2 is a cell surface protein involved in the control of blood pressure.

It is found on lung, heart, kidney, and intestinal cells

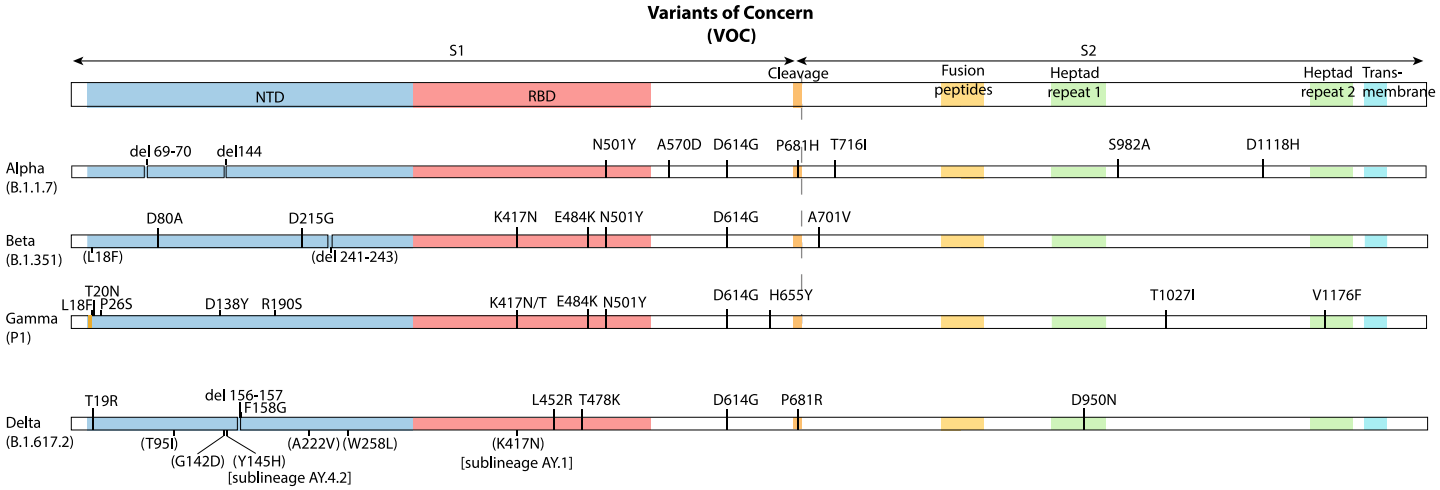
Spike mutations change how well the virus infects cells



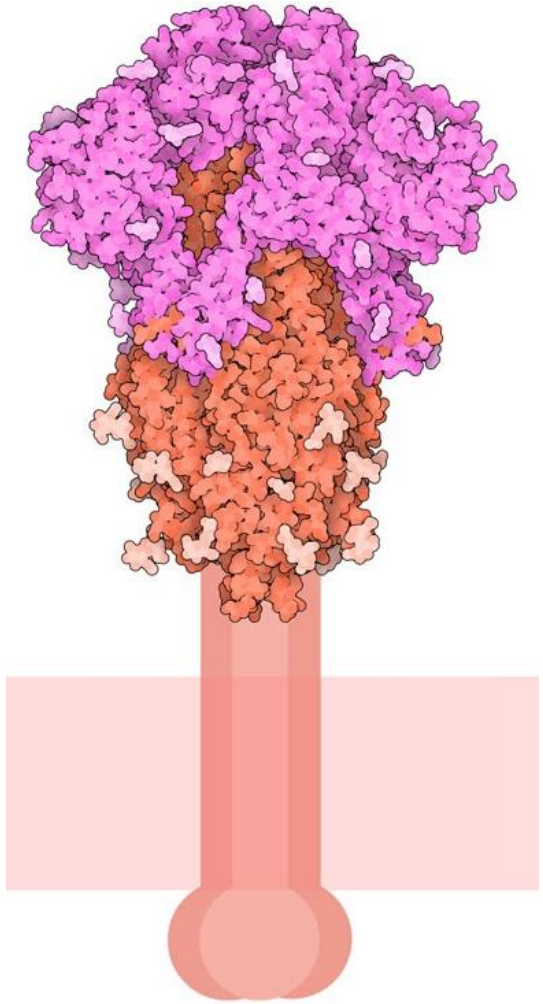
SARS-CoV-2 **Spike**



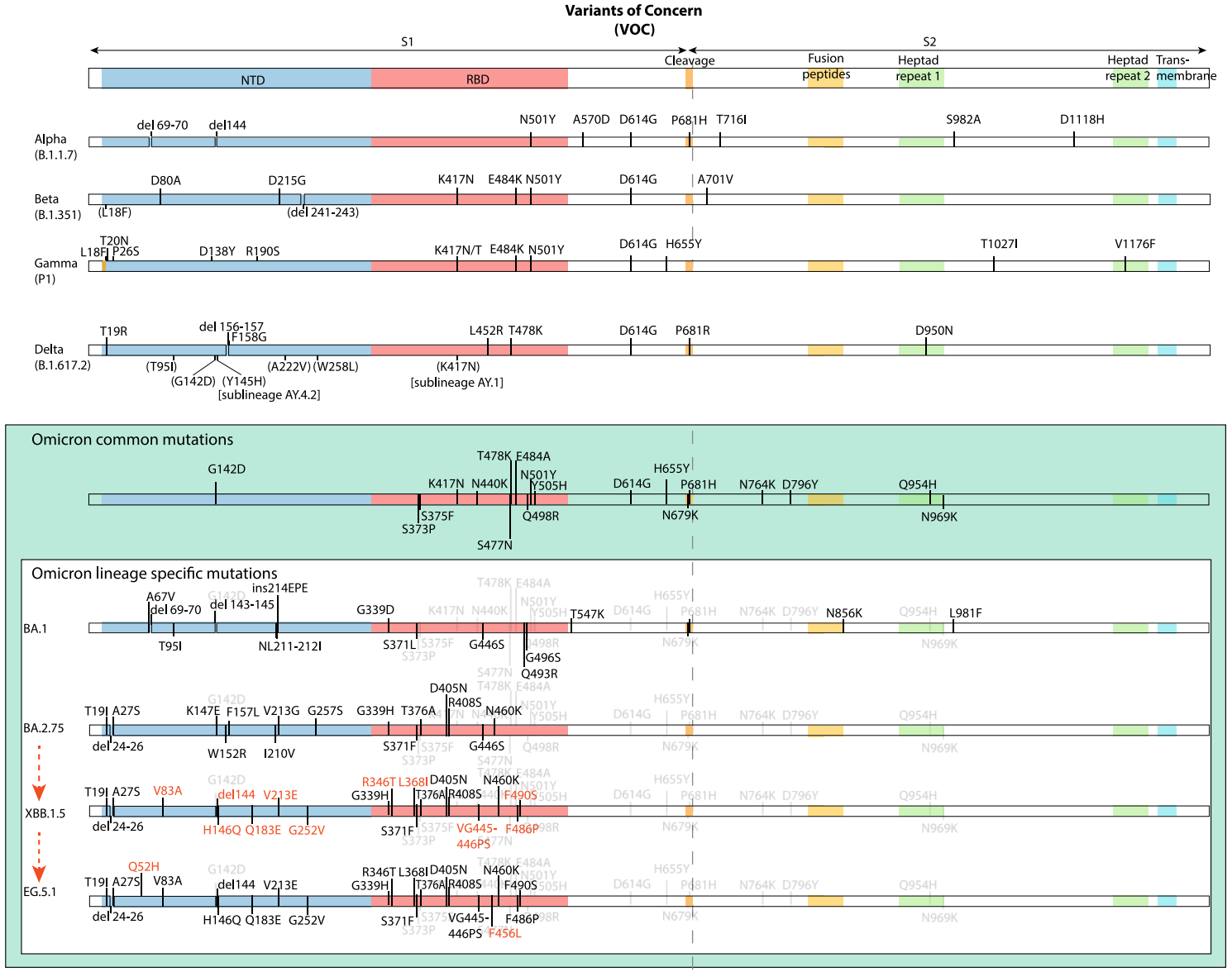
Spike mutations change how well the virus infects cells



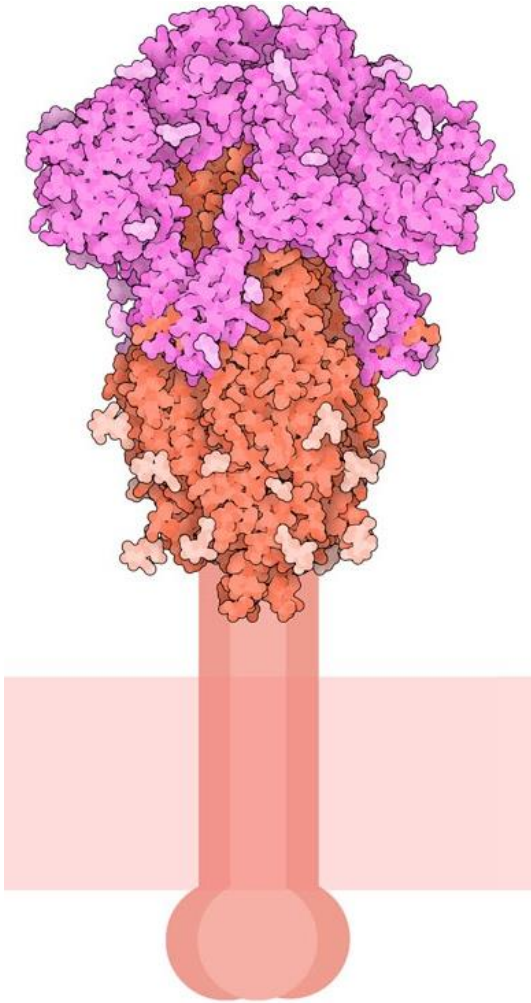
SARS-CoV-2 **Spike**



Spike mutations change how well the virus infects cells



SARS-CoV-2 **Spike**



| Virology | Announcement | 3 February 2022

Identification of the First SARS-CoV-2 Lineage B.1.1.529 Virus Detected in Europe

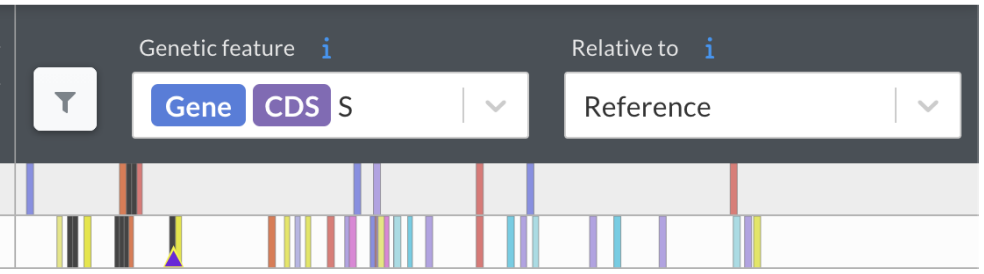
Authors: Bert Vanmechelen, Anne-Sophie Logist, Tony Wawina-Bokalanga, Jens Verlinden, Joan Martí-Carreras, Caspar Geenen, Bram Slechten, Lize Cuypers, Emmanuel André, Guy Baele, Piet Maes | AUTHORS INFO & AFFILIATIONS

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Pango lineage (Nextclade)	WHO name	Mut.	non-ACGTN	Ns	Cov.	Gaps	Ins.	FS	SC
B.1.617.2	Delta	36	0	0	99.8%	13	0	0	0
BA.1	Omicron	54	0	0	99.4%	39	9	0	0



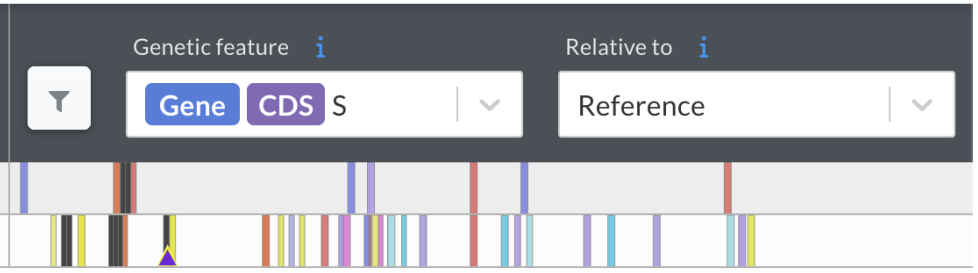
*variant comparison using nextclade

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Authors: Bert Vanmechelen, Anne-Sophie Logist, Tony Wawina-Bokalanga, Jens Verlinden, Joan Martí-Carreras, Caspar Geenen, Bram Slechten, Lize Cuypers, Emmanuel André, Guy Baele, Piet Maes | AUTHORS INFO & AFFILIATIONS

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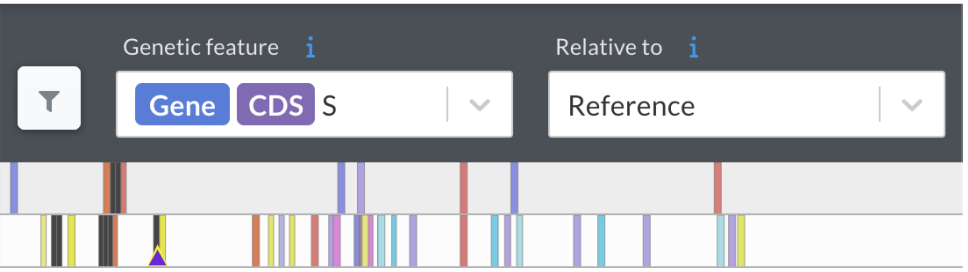
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BUT
Negative for SARS-CoV-2 Spike

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Further investigation and identification of novel variant!

Quickly responding and adapting to ongoing epidemics is only possible with open knowledge and data







